

Gaussian Elimination: Math Club Notes

Markus Hoehn

Linear Systems

Simple System

$$3x_1 + 2x_2 = -6$$

$$2x_1 + 5x_2 = 7$$

To solve this simple system, we'll use the substitution method.

Linear Systems

Simple System Solution

Solution: $(-4, 3)$

Linear Systems

Another Simple System

$$x_1 + 2x_2 - x_3 = 2$$

$$2x_1 - 3x_2 + x_3 = -1$$

$$5x_1 - x_2 - 2x_3 = -3$$

In this slightly more complex system, we can still employ the substitution method to determine the values of x_1 , x_2 , and x_3 .

Linear Systems

Another Simple System Solution

Solution: $(1, 2, 3)$

Linear Systems

Harder System

$$x_1 + 2x_2 - 3x_3 + 4x_4 = 12$$

$$2x_1 + 2x_2 - 2x_3 + 3x_4 = 10$$

$$x_2 + x_3 = -1$$

$$x_1 - x_2 + x_3 - 2x_4 = -4$$

Solving this harder system using substitution is extremely time-consuming. For larger systems, like 50 equations with 50 variables, it becomes impractical. In many STEM fields that require computational work, linear systems like these are common, often much larger in scale. You may have thousands or millions of linear equations. Now you could attempt to write code for our substitution method, but this again becomes impractical.

Intro

Thankfully, computers come equipped with powerful GPUs specifically designed for tasks such as matrix multiplication and other matrix operations. We will want to harness that power. To do this, we will look at our topic, Gaussian Elimination, a method for solving systems of linear equations using matrices.

Matrices

Matrices

In its most basic computational form, a matrix is a rectangular array of numbers. A matrix with n rows and m columns is denoted as an $n \times m$ matrix. Here's an example of a 2×3 matrix:

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 12 & 11 & 7 \end{bmatrix}$$

Matrices

Common Types of Matrices

Here are some common types of matrices:

1. **Row Matrix** ($n = 1$):

$$\mathbf{v} = [1 \ 2 \ 3]$$

2. **Column Matrix** ($m = 1$):

$$\mathbf{w} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

3. **Square Matrix** ($n = m$):

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 2 \\ 4 & 5 & 0 \end{bmatrix}$$

In this case, B is a 3×3 square matrix.

Matrix Multiplication

Matrix multiplication is defined for multiplying an $n \times m$ matrix by an $m \times p$ matrix and gives an $n \times p$ matrix.

Matrix Multiplication

Row and Column Matrices

$$\mathbf{A} = [1, 3, 4, 6]$$

$$\mathbf{B} = \begin{bmatrix} 5 \\ 9 \\ 8 \\ 7 \end{bmatrix}$$

$$\mathbf{A} \cdot \mathbf{B} = 5 \cdot 1 + 9 \cdot 3 + 8 \cdot 4 + 7 \cdot 6 = 106$$

Matrix Multiplication

General Case

Matrix multiplication can be seen as a combination of these row and column multiplications. Consider the matrices:

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 3 & 1 & 4 \\ -2 & 2 & 3 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 5 & -1 & 2 \\ 0 & 3 & -2 \\ 1 & 2 & -3 \end{bmatrix}$$

The element at the intersection of the first row and first column in our product matrix $\mathbf{A} \cdot \mathbf{B}$ is obtained by multiplying the corresponding elements from the first row of $\mathbf{A}$ with the elements in the first column of $\mathbf{B}$, and this process extends to all elements in the resulting matrix.

Now multiply the matrices.

Matrix Multiplication

General Case Solution

Solution:

$$\mathbf{A} \cdot \mathbf{B} = \begin{bmatrix} 9 & -4 & 7 \\ 19 & 8 & -8 \\ -7 & 14 & -17 \end{bmatrix}$$

Augmented Matrix Form

Consider the 4×4 matrix below:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 2 & 2 & -2 & 3 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -2 \end{bmatrix}$$

Also consider the 4×1 matrix below:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

We can multiply our matrices since the number of columns of $\mathbf{A}$ is the same as the number of rows of $\mathbf{x}$.

$$\mathbf{Ax} = \begin{bmatrix} x_1 + 2x_2 - 3x_3 + 4x_4 \\ 2x_1 + 2x_2 - 2x_3 + 3x_4 \\ x_2 + x_3 \\ x_1 - x_2 + x_3 - 2x_4 \end{bmatrix}$$

Augmented Matrix Form

The observant listener may notice these linear expressions look like the left-hand side of our harder linear system. We can convert our linear system problem to matrix language by adding the right-hand side as a matrix $\mathbf{b}$ and asking when $\mathbf{Ax} = \mathbf{b}$.

$$\mathbf{b} = \begin{bmatrix} 12 \\ 10 \\ -1 \\ -4 \end{bmatrix}$$

Putting it together nicely, we have

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 2 & 2 & -2 & 3 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 12 \\ 10 \\ -1 \\ -4 \end{bmatrix}$$

Augmented Matrix Form

We can write this matrix equation more concisely using the notation of an augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 2 & 2 & -2 & 3 & 10 \\ 0 & 1 & 1 & 0 & -1 \\ 1 & -1 & 1 & -2 & -4 \end{array} \right]$$

We have successfully transferred our linear systems process to the language of matrices. Now how will we use this?

Gaussian Elimination

Elementary Row Operations

There are three types of elementary row operations we can apply to our augmented matrix:

1. Swapping two rows
2. Multiplying a row by a nonzero number
3. Adding a multiple of one row to another row

I am going to claim that performing any of these row operations on our augmented matrix will not alter the solution. To demonstrate this, we can step back to our previous language where "rows" of our augmented matrix are equations.

Gaussian Elimination

Elementary Row Operations: Swapping

Consider the following system of equations:

$$2x_1 + 3x_2 = 7$$

$$4x_1 - 2x_2 = 2$$

It's evident that this system has an equivalent solution when the order of the equations is swapped:

$$4x_1 - 2x_2 = 2$$

$$2x_1 + 3x_2 = 7$$

Gaussian Elimination

Elementary Row Operations: Multiplying a row by a nonzero scalar

Now, let's delve into the second operation. Consider the solutions of:

$$\begin{aligned}2(2x_1 + 3x_2) &= 2(7) \\ 4x_1 - 2x_2 &= 2\end{aligned}$$

Again, it's evident that this operation doesn't alter the solutions, drawing upon our intuition from middle school algebra.

Gaussian Elimination

Elementary Row Operations: Adding a multiple of one row to another row

The final case is a bit more intricate. Why doesn't adding a multiple of one equation to another equation change the solution set? Let's explore this:

$$\begin{aligned}2x_1 + 3x_2 + k(4x_1 - 2x_2) &= 7 + k(2) \\4x_1 - 2x_2 &= 2\end{aligned}$$

Although this appears different, since $4x_1 - 2x_2 = 2$, we can employ substitution to rewrite it as follows:

$$\begin{aligned}2x_1 + 3x_2 + k(2) &= 7 + k(2) \\4x_1 - 2x_2 &= 2\end{aligned}$$

Now, we can confidently conclude that this is valid. We've essentially added the same constant to both sides of the equation, relying on our intuition from middle school algebra.

Now that we believe we can perform row operations without changing our solutions, how does this help?

Gaussian Elimination

Row Reduction

$$\left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 2 & 2 & -2 & 3 & 10 \\ 0 & 1 & 1 & 0 & -1 \\ 1 & -1 & 1 & -2 & -4 \end{array} \right] \xrightarrow{\substack{R_2 - 2R_1 \\ R_4 - R_1}} \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 0 & -2 & 4 & -5 & -14 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & -3 & 4 & -6 & -16 \end{array} \right]$$

$$\xrightarrow{R_2 / -2} \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 0 & 1 & -2 & \frac{5}{2} & 7 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & -3 & 4 & -6 & -16 \end{array} \right] \xrightarrow{\substack{R_1 - 2R_2 \\ R_3 - R_2 \\ R_4 + 3R_2}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -2 \\ 0 & 1 & -2 & \frac{5}{2} & 7 \\ 0 & 0 & 3 & -\frac{5}{2} & -8 \\ 0 & 0 & -2 & \frac{3}{2} & 5 \end{array} \right]$$

$$\xrightarrow{R_3 / 3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & -2 \\ 0 & 1 & -2 & \frac{5}{2} & 7 \\ 0 & 0 & 1 & -\frac{5}{6} & -\frac{8}{3} \\ 0 & 0 & -2 & \frac{3}{2} & 5 \end{array} \right] \xrightarrow{\substack{R_1 - R_3 \\ R_2 + 2R_3 \\ R_4 + 2R_3}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{6} & \frac{2}{3} \\ 0 & 1 & 0 & \frac{5}{6} & \frac{10}{3} \\ 0 & 0 & 1 & -\frac{5}{6} & -\frac{8}{3} \\ 0 & 0 & 0 & -\frac{1}{6} & -\frac{14}{3} \end{array} \right]$$

$$\xrightarrow{-6R_4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{6} & \frac{2}{3} \\ 0 & 1 & 0 & \frac{5}{6} & \frac{10}{3} \\ 0 & 0 & 1 & -\frac{5}{6} & -\frac{8}{3} \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{\substack{R_1 + \frac{1}{6}R_4 \\ R_2 - \frac{5}{6}R_4 \\ R_3 + \frac{5}{6}R_4}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

Gaussian Elimination

Row Reduction

If you found that to be painful, substitution is far worse. You're welcome to give it a try. This is where you'll truly understand why Gaussian Elimination earned its name. The objective of the row reduction process is to eliminate all values outside the diagonal and transform the remaining values into 1. This gives a nice correspondence between our $\mathbf{x}$ matrix and $\mathbf{b}$ matrix. Putting our augmented matrix back into $\mathbf{Ax} = \mathbf{b}$ form:

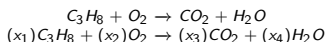
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix} \implies \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}$$

Thus, we have our solution. Now let's apply our knowledge.

Stoichiometry

The Problem and Setup

We could use our knowledge to quickly balance chemical equations. On both sides of this combustion reaction of propane, we want the same number of carbon, oxygen, and hydrogen atoms, so we need to figure out the stoichiometric coefficients.



$$\begin{bmatrix} \text{Number of Carbon Atoms} \\ \text{Number of Hydrogen Atoms} \\ \text{Number of Oxygen Atoms} \end{bmatrix}$$

$$\begin{aligned} x_1 \begin{bmatrix} 3 \\ 8 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} &= x_3 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \\ \Rightarrow x_1 \begin{bmatrix} 3 \\ 8 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} - x_3 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - x_4 \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ = \begin{bmatrix} 3 & 0 & -1 & 0 \\ 8 & 0 & 0 & -2 \\ 0 & 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

Stoichiometry

Row Reduction

As an augmented matrix:

$$\begin{aligned} & \left[\begin{array}{cccc|c} 3 & 0 & -1 & 0 & 0 \\ 8 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right] \xrightarrow{R_1/3} \left[\begin{array}{cccc|c} 1 & 0 & -\frac{1}{3} & 0 & 0 \\ 8 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right] \\ & \xrightarrow{R_2 - 8R_1} \left[\begin{array}{cccc|c} 1 & 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{8}{3} & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 2 & -2 & -1 & 0 \\ 0 & 0 & \frac{8}{3} & -2 & 0 \end{array} \right] \\ & \xrightarrow{\substack{R_2/2 \\ \frac{3}{8}R_3}} \left[\begin{array}{cccc|c} 1 & 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{3}{4} & 0 \end{array} \right] \xrightarrow{\substack{R_1 + \frac{1}{3}R_3 \\ R_2 + R_3}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 1 & 0 & -\frac{5}{4} & 0 \\ 0 & 0 & 1 & -\frac{3}{4} & 0 \end{array} \right] \end{aligned}$$

Notice that x_4 is a free variable that the other variables depend on. Let $x_4 = t$.

$$x_1 - \frac{1}{4}t = 0 \implies x_1 = \frac{1}{4}t$$

$$x_2 - \frac{5}{4}t = 0 \implies x_2 = \frac{5}{4}t$$

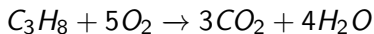
$$x_3 - \frac{3}{4}t = 0 \implies x_3 = \frac{3}{4}t$$

Stoichiometry

It is logical to have a single free variable because when we proportionally scale all the coefficients, the balance of our equation should remain unaffected.

However, we want the lowest value of t such that every value is an integer, so we choose the solution where $t = 4$.

Thus, we have $x_1 = 1, x_2 = 5, x_3 = 3, x_4 = 4$.



and our reaction is balanced.